Nalanda University

Nalanda was an acclaimed Mahavihara, a large Buddhist monastery in the ancient kingdom of Magadha in India. The site is located about ninety five kilometres southeast of Patna near the town of Bihar Sharif and was a centre of learning from the fifth century CE to c. 1200 CE. It is a UNESCO World Heritage Site.

The highly formalised methods of Vedic learning helped inspire the establishment of large teaching institutions such as Taxila, Nalanda and Vikramashila, which are often characterised as India's early universities. Nalanda flourished under the patronage of the Gupta Empire in the fifth and sixth centuries and later under Harsha, the emperor of Kanauj. The liberal cultural traditions inherited from the Gupta age resulted in a period of growth and prosperity until the ninth century. The subsequent centuries were a time of gradual decline, a period during which Buddhism became popular in eastern India patronised by the Palas of Bengal.



Nalanda University

At its peak, the Nalanda attracted scholars and students from near and far with some travelling all the way from Tibet, China, Korea and Central Asia. Archaeological findings also confirm the contact with the Shailendra dynasty of Indonesia, one of whose kings built a monastery in the complex.

Nalanda was ransacked and destroyed by an army of the Mamluk dynasty of the Delhi Sultanate under Bakhtiyar Khalji in c. 1200 CE. While some sources note that the Mahavihara continued to function in a makeshift fashion for a little longer, it was eventually abandoned and forgotten. The site was accidentally discovered when the Archaeological Survey of India surveyed the area. Systematic excavations commenced in 1915, which unearthed 11 monasteries and 6 brick temples situated on 12 hectares (30 acres) of land. A trove of sculptures, coins, seals and inscriptions have also been discovered since then and all of them are on display in the Nalanda Archaeological Museum situated nearby. Nalanda is now a notable tourist destination and a part of the Buddhist tourism circuit. Recently, the government of India, in cooperation with other South and South-east Asian countries, has revived this university.

Gupta Sciences

Mathematics and Astronomy

The invention of the theory of zero and the consequent evolution of the decimal system are to be credited to the thinkers of this age. In the *Surya Siddanta*, Aryabhatta (belonging to late fifth and early sixth century CE) examined the true cause of the solar eclipses. In calculation of the size of the earth, he is very close to the modern estimation. He was the first astronomer to discover that the earth rotates on its own axis. He is also the author of *Aryabhattiyam*, which deals with arithmetic, geometry and algebra.

Varahamihira's *Brihat Samhita* (sixth century CE) is an encyclopaedia of astronomy, physical geography, botany and natural history. His other works are *Panch Siddhantika* and *Brihat Jataka*. Brahmagupta (late sixth and early seventh century CE) is author of important works on mathematics and astronomy, namely *Brahmasphuta-siddhanta* and *Khandakhadyaka*.

Medical Sciences

Metallic preparations for the purpose of medicine and references to the use of mercury and iron by Varahamihira and others indicate that much progress was made in chemistry. The *Navanitakam* was a medical work, which is a

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